

Beam Failure Prediction

Predict where a beam will fail, test it, and explain the difference between prediction and evidence.

Intermediate	1 class period	Practice & Review
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Engineering Context

A rover instrument bracket uses a cantilevered support. Before destructive testing, engineers must predict the critical region and expected failure mode.

What You Will Practice

- Draw free-body diagrams
- Identify bending moment trends
- Distinguish stress from deflection
- Compare prediction with test evidence

What You Need

- Uniform beam samples
- Clamp or support fixture
- Incremental masses
- Ruler
- Marker
- Engineering notebook

Student Instructions

- 1 Measure the beam geometry and mark several observation stations.
- 2 Draw the beam, support, applied load, and reactions.
- 3 Predict the region of maximum bending moment and likely failure location.
- 4 Predict whether the beam will bend elastically, permanently deform, buckle, or fracture.
- 5 Load the beam in approved increments and record deflection at one location.
- 6 Stop at the instructor limit or after failure.
- 7 Compare the observed behavior with the prediction.
- 8 Write one reason for any mismatch.

Engineering Requirements

- Complete prediction before testing
- Use consistent load position
- Record load and deflection for every increment
- Identify observed failure mode with evidence

Constraints & Safety

- Wear eye protection
- Use a shield for brittle samples if directed
- Keep body parts away from the failure zone
- Do not exceed approved loads

What You Submit

- Free-body diagram
- Prediction statement
- Load-deflection table
- Failure photo or sketch
- Prediction-versus-result analysis

Reflection Questions

- 1 Why was the support region critical?
- 2 When did behavior stop appearing elastic?
- 3 What measurement uncertainty affected the graph?
- 4 How would doubling thickness change stiffness?

Engineering Tip

Mark the beam before testing so you can connect deformation to a specific location.

Common Mistake

Predicting failure only from where the load is applied ignores the internal bending moment distribution.

Stretch Goal

Compare two beams with equal mass but different cross-section shapes.

Career Connection

Structural test engineers compare analytical predictions with physical tests before certifying flight hardware.